Supervised ML Algo < Regression / Classificⁿ

Linear Regression < Simple / Multiple / Polynomial

Simple LR - 1 input & 1 output
↳ plot (sort of linear) data - Draw best-fit line
↳ minimises error for all pts.

$y = mx + b$

- Import lib & dataframe. Split into train & test.
- Train data & it finds best-fit line i.e m & b (m ↓ coef-, b ↓ inter cept-)
- Test (predict using model)

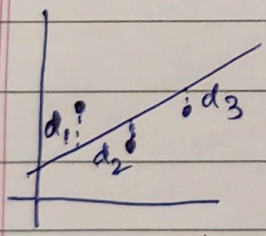
m → weightage - dependence on x
b → offset - initial y even when x = 0

Finding m, b - < closed form - direct formula / non- " " - apxⁿ techniques

$b = \bar{y} - m\bar{x}$

$$m = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2}$$

$\bar{x}, \bar{y}$ ↓ mean



$$\text{Error} = \sum_{i=1}^{n} d_i^2$$ → Sq to eliminate +ve & -ve

$$\text{Error} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$ ← $d_i = y_i - \hat{y}_i$ → predicted val
↳ actual val. acc to data

$$= \sum_{i=1}^{n} (y_i - mx_i - b)^2 \text{ as } (\hat{y}_i = mx_i + b)$$

If $b=0$, Error $= \sum_{i=1}^{n} (y_i - mx_i)^2$. So change only m. (line at (0,0)).

As we move line, a line wud have min error at 'm'.

If $m=1$, $\varepsilon = \sum_{i=1}^{n} (y_i - x_i - b)^2$, change only b.

As we move line ↑↓, a line min error at 'b'.

Merge both, a value of m & b where min error.

$$\frac{\partial \varepsilon}{\partial m} = 0 \quad \& \quad \frac{\partial \varepsilon}{\partial b} = 0 \rightarrow b = \bar{y} - m\bar{x}$$

$$\downarrow$$

$$m = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2}$$